SYNTHETIC VALIDATION − SuSiE versus EM SuSiE−mix | iteration 18

650 eligible gene−tissue regions across 5 tissues | tentative four−panel comparison

A Where do CS counts agree?

Rows: SuSiE CSs | Columns: EM CSs | Cells: gene−tissue regions

Adipose Tissue

Most disagreement | n = 130
Same: 57 (43.8%) | Different: 73

	0	1	2	3	4	5	6	7	8
0	−	10	0	1	1	3	1	0	1
1	2	10	0	1	3	1	0	1	1
2	1	0	7	0	1	0	0	1	0
3	1	0	0	4	2	0	3	0	1
4	0	4	1	2	6	2	1	3	0
5	1	1	1	0	1	9	0	0	0
6	2	0	0	1	1	1	6	0	1
7	0	0	0	1	0	0	1	10	0
8	2	2	2	0	1	3	0	2	5

Blood Vessel

Most disagreement | n = 130
Same: 58 (44.6%) | Different: 72

	0	1	2	3	4	5	6	7	8
0	−	7	0	1	1	1	0	1	0
1	1	12	3	1	0	2	0	0	0
2	1	0	9	0	3	2	2	0	2
3	0	3	0	7	0	0	0	1	2
4	0	0	0	1	3	0	1	1	3
5	0	1	1	1	1	10	1	0	0
6	2	0	1	2	0	1	8	2	0
7	2	1	2	2	1	1	0	5	3
8	1	3	1	1	0	1	0	0	4

Salivary Gland

Most agreement | n = 130
Same: 101 (77.7%) | Different: 29

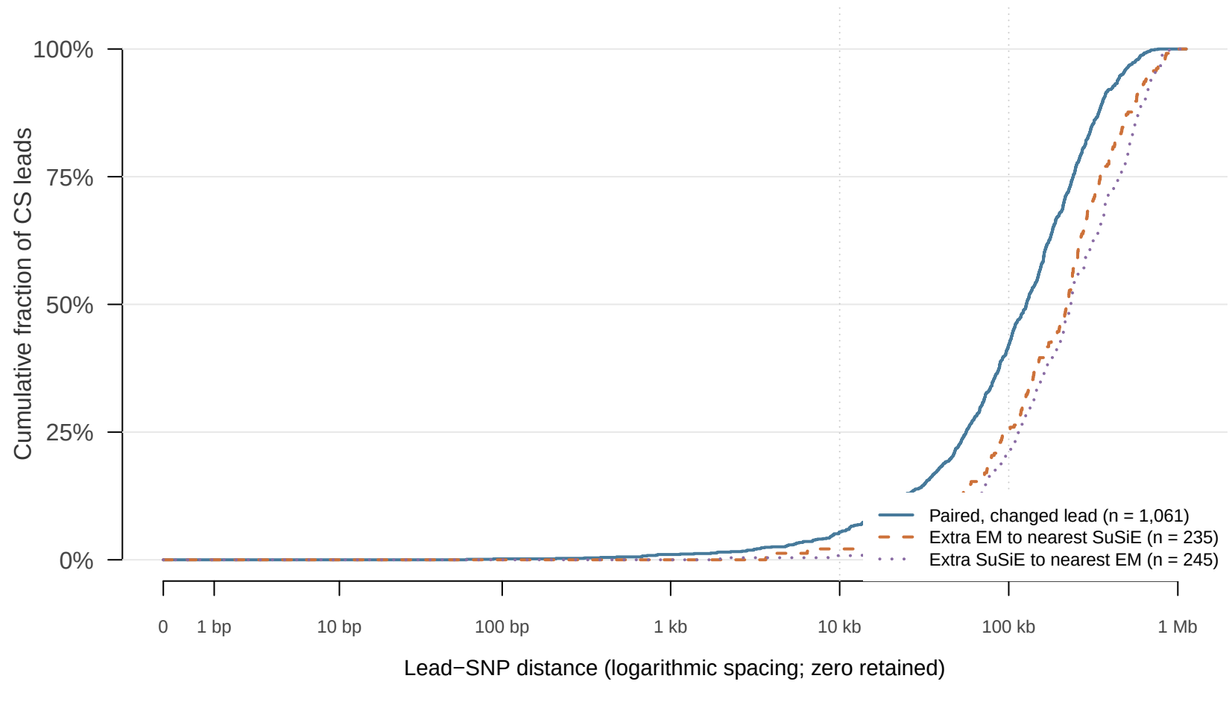
	0	1	2	3	4	5	6	7	8
0	−	16	0	0	0	1	0	0	0
1	0	10	0	0	0	0	0	0	0
2	0	0	12	1	0	0	1	2	0
3	0	0	0	19	0	0	0	0	0
4	0	0	0	0	12	0	0	0	1
5	0	0	1	0	1	11	0	0	1
6	0	1	0	1	0	2	14	0	0
7	0	0	0	0	0	0	0	12	0
8	0	0	0	0	0	0	0	0	11

Small Intestine

Most agreement | n = 130
Same: 96 (73.8%) | Different: 34

	0	1	2	3	4	5	6	7	8
0	−	14	0	0	0	0	0	0	0
1	0	16	0	2	0	0	0	0	2
2	0	0	10	0	1	1	0	0	0
3	0	1	0	13	0	0	0	0	0
4	0	0	1	0	8	0	1	1	0
5	0	0	0	0	0	18	0	0	0
6	1	0	0	0	0	0	13	0	0
7	3	2	0	1	0	0	0	11	0
8	0	0	1	1	0	0	0	1	7

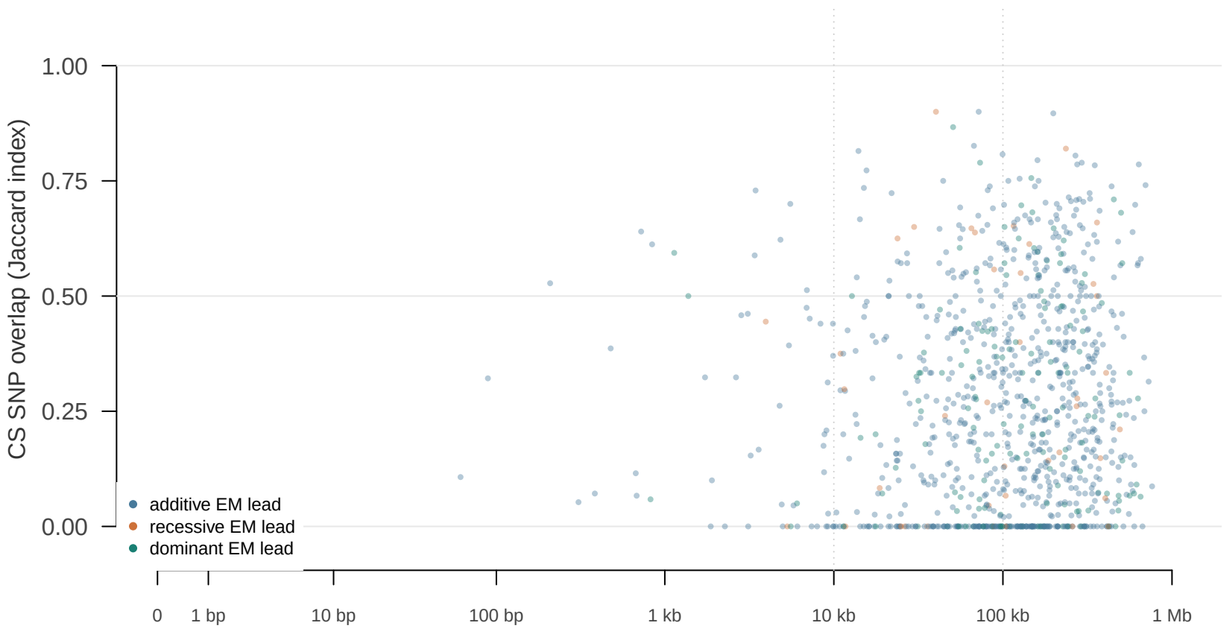
B Lead retention and displacement: all tissues



Not in curves: 1,112 identical pairs; 262 extras with no other−model CS

C Does a shifted lead change the credible set?

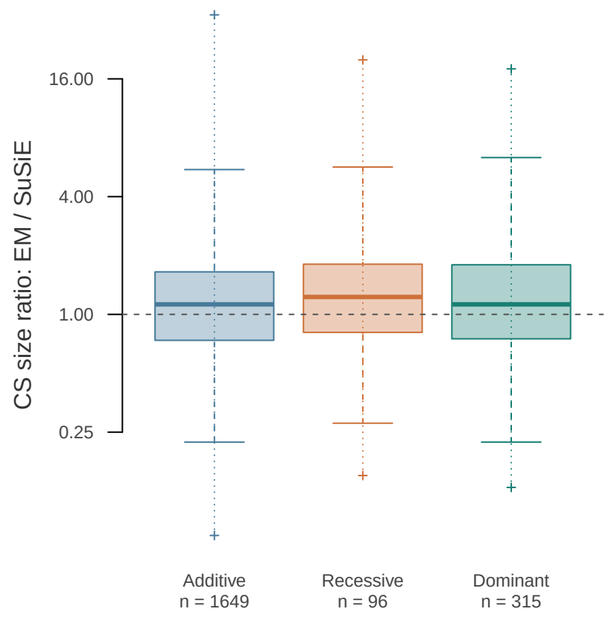
Changed pairs: 1,003 | identical sets: 0 | disjoint sets: 192



Jaccard: 0 = disjoint; 1 = identical SNP sets. Missing membership: 58 pairs.

D Resolution and support

Matched CSs, grouped by EM lead coding

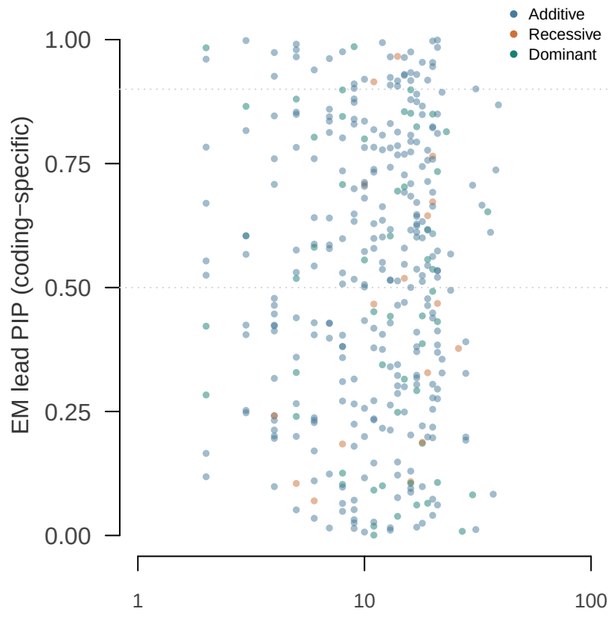


Below 1: fewer SNPs in the EM CS

Missing size: 113 pairs | whiskers: 1.5 IQR; +: range

Additional EM CSs

n = 362 | PIP >= 0.9: 39



Unavailable size/PIP: 13; no SuSiE CS: 140

SNP sets collapse genetic codings. Pairing does not establish biological equivalence; smaller sets do not establish better coverage.

Pairing reserves identical lead IDs, then minimizes total distance without reusing CSs. Additional = unmatched under this rule.

Cohort: P < 1e−08; mean reads >= 100; at least one CS in either model. Tissue ranking requires n >= 100.